10 t-distribution

Recall a 95% z-confidence interval for the population mean parameter is:

$$z = \frac{\bar{x} - \mu}{\hat{\sigma}/\sqrt{n}}$$

not normal

$$(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}})$$

$$\bar{x} \pm 1.96\, \sigma/\sqrt{n}$$

However we have assumed that we know σ. This interval cannot be used for data analysis because typically we do not know σ.

How about we replace σ by the estimate: $\hat{\sigma}$=sample SD?

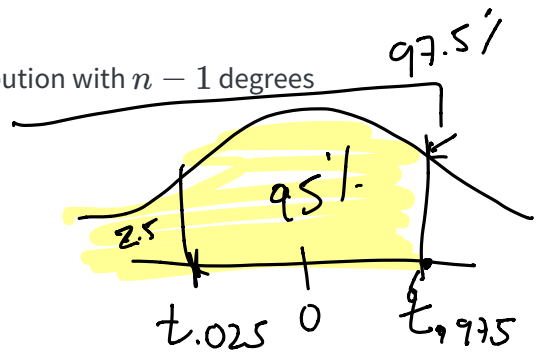
When the data is normally (Gaussian) distributed and we replace σ by $\hat{\sigma}$ =sample SD, then the corresponding 95% confidence interval is:

$$(\bar{x} - t_{.975}\frac{\hat{\sigma}}{\sqrt{n}}, \bar{x} + t_{.975}\frac{\hat{\sigma}}{\sqrt{n}})$$

$$\bar{x} \pm t_{.975}\, \hat{\sigma}/\sqrt{n}$$

.

wher $t_{.975}$ is the 97.5 percentile of the central t-distribution with $n-1$ degrees of freedom.



or equivalently:

$$\bar{x} \pm t_{.975}\frac{\hat{\sigma}}{\sqrt{n}}$$

. or

$$\bar{x} \pm t_{.025}\frac{\hat{\sigma}}{\sqrt{n}}$$

$$\bar{x} \pm t_{.025}\, \hat{\sigma}/\sqrt{n}$$

.

where $t_{.025}$ is the 2.5 percentile of the central $t-$distribution with $n-1$ degrees of freedom (df).

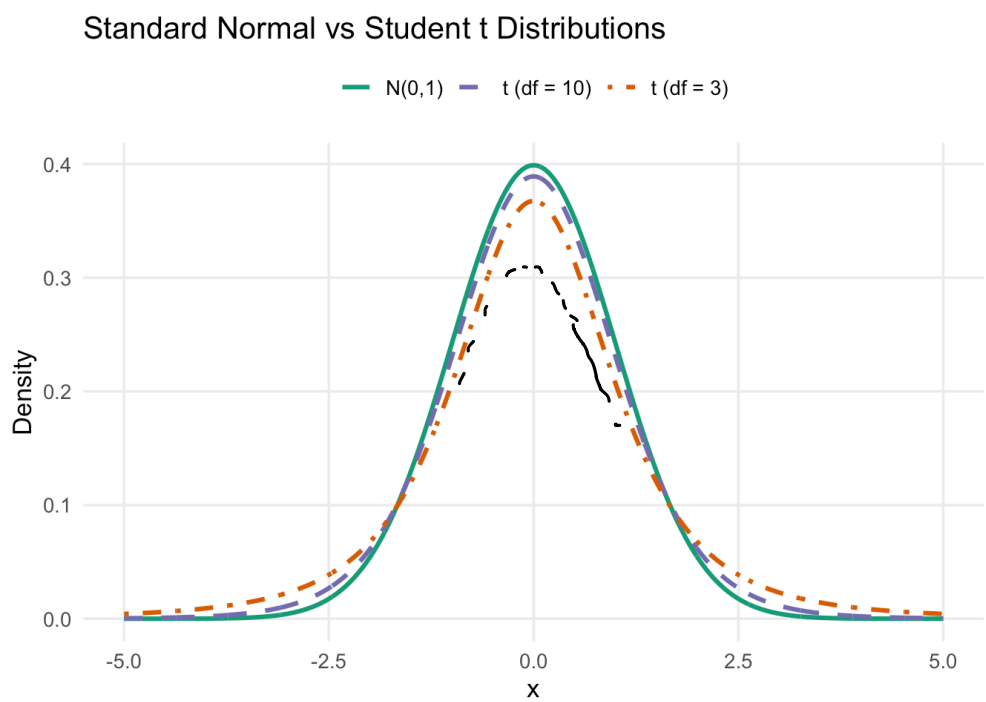
The following plots compare the central t-distribution densities with different degrees of freedom to the standard normal distribution. As the degrees of freedom increases, the t-distribution gets closer and closer to the standard normal distribution.

```
# Plot: Standard Normal vs t(df=3) and t(df=10)
library(ggplot2)

x <- seq(-5, 5, length.out = 2000)
```

```r
df <- data.frame(
  x = rep(x, 3),
  density = c(dnorm(x), dt(x, df = 3), dt(x, df = 10)),
  dist = factor(rep(c("N(0,1)", "t (df = 3)", "t (df = 10)"), ea
                levels = c("N(0,1)", "t (df = 10)", "t (df = 3)"
)

ggplot(df, aes(x = x, y = density, color = dist, linetype = dist
  geom_line(linewidth = 1.1) +
  scale_color_manual(values = c("N(0,1)" = "#1b9e77",
                                "t (df = 10)" = "#7570b3",
                                "t (df = 3)"  = "#d95f02")) +
  scale_linetype_manual(values = c("N(0,1)" = "solid",
                                   "t (df = 10)" = "dashed",
                                   "t (df = 3)"  = "dotdash")) +
  labs(title = "Standard Normal vs Student t Distributions",
       x = "x", y = "Density", color = NULL, linetype = NULL) +
  coord_cartesian(xlim = c(-5, 5)) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "top",
        panel.grid.minor = element_blank())
```



Recall the quiz 3 problem:

Example: Monarch butterflies

Assume the population of monarch butterflies in Natural Bridges State Beach has a normally distributed wingspan. Assume also that you collect a sample of 100 butterflies and compute an average wingspan of 9.6 cm and estimate the sample standard deviation to be 0.3. What is a 95% confidence interval for the population mean?

n-1

97.5

$t_{.975} = 1.98$

```
t975<-qt(.975,df=99)
t975
```

```
[1] 1.984217
```

$\hat{\sigma} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

```
se<-0.3/sqrt(100)
9.6-t975*se
```

```
[1] 9.540473
```

```
9.6+t975*se
```

```
[1] 9.659527
```

We are 95% confident the the average wingspan is within (9.54,9.65).

What happens if our population is not normally distributed? When can I use the CLT and normal approximation?

If the population is not Gaussian, the t-interval is often still approximately correct when:

- n is moderately large (CLT $\bar{X}$ kicks in), and
- The distribution isn't extremely skewed
- Observations are i.i.d. (independent and identically distributed) with finite variance.

$$\int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 0.95$$

$$h = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad .9$$